

# Sairam Sri Vatsavai

Assistant Computational Scientist, Brookhaven National Laboratory

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Google Scholar: [Scholar Profile](#) GitHub: [Sairam954](#)

## Research Interests

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Analytical and AI-enabled modeling of emerging computing technologies (photonic, superconducting digital, and in-memory computing) across the device, architecture, and application layers; cross-layer design space exploration and hardware-software co-design for large-scale AI workloads; modeling, simulation, and learned surrogates for large-scale distributed computing systems; workload modeling, scheduling, and resource management for resilient scientific workflows; agentic AI systems for scientific user facilities.

## Education

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### Ph.D. in Electrical Engineering

Aug 2019 – May 2024

University of Kentucky, Lexington, KY

GPA: 4.0/4.0

*Dissertation:* Cross-Layer Design of Highly Scalable and Energy-Efficient AI Accelerator Systems Using Photonic Integrated Circuits

Advisor: Dr. Ishan Thakkar

### B.Tech. in Electronics and Communication Engineering

Aug 2012 – May 2016

Jawaharlal Nehru Technological University, India

GPA: 3.75/4.0

## Professional Experience

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### Brookhaven National Laboratory, Upton, NY

Computing and Data Sciences Department – Systems, Architectures, and Advanced Technologies Group

*Research Staff 3 / Assistant Computational Scientist*

May 2026 – Present

*Postdoctoral Research Associate*

May 2024 – May 2026

- **REDWOOD (DOE ASCR):** Architected **CGSim**, a scalable SimGrid-based discrete-event simulation framework for large-scale distributed computing infrastructures, achieving near-linear scalability across tens of sites and thousands of concurrent jobs with real-time visualization and event-level ML training data generation (Best Short Paper, PMBS @ SC'25; IEEE eScience 2026)
- Developed **TRASSE**, a transformer-based learned simulator surrogate for scheduling-policy search, with per-site training pipelines and autoregressive rollout across **74 WLCG sites**, achieving 88.0% job-selection accuracy, 99.96% state-classification accuracy, and a 2.6-minute median job-duration error
- Extending CGSim beyond the ATLAS/WLCG grid to model computing infrastructure for the **Vera C. Rubin Observatory**
- Developed **HOLOS**, a gradient-descent-based design space exploration framework for hierarchical AI accelerator systems that jointly searches parallelism strategy (data, tensor, pipeline, microbatching) and accelerator geometry across CMOS and superconducting digital technologies, with analytical latency, energy, and area models; achieves orders-of-magnitude speedup over exhaustive, random, Bayesian, and genetic search
- Performed **system-level exploration of superconducting supercomputers** for emerging AI workloads, modeling LLM training and inference with validated correlation to production systems and industry tools (Calculon)
- Lead developer of **LightshowAI**, an X-ray absorption spectroscopy platform for NSLS-II; integrated BNL SSO/OIDC federated authentication, Tiled beamline data streaming, and a Chainlit chatbot in-

terface, and connected the platform to DOE American Science Cloud (AmSC) model services under the Genesis Mission

- Designed **LightshowAI-Agent**, an MCP-based agentic framework exposing every capability as a tool (three MCP servers, 32 tools) spanning Materials Project, a facility-hosted GNN XANES surrogate, and AmSC MLflow model services with provenance tracking; live demonstration at Confab26 featured on the AmSC website
- **Co-PI of COSMOS** (BNL LDRD), leading benchmarking and evaluation of multidimensional photonic computing systems against GPU, TPU, and single-degree-of-freedom photonic baselines
- Established **GitLab CI/CD pipelines** with Kaniko-based containerized builds and automated code validation for research software deployment

## Graduate Research Assistant

Aug 2019 – May 2024

University of Kentucky, Lexington, KY

UCAT Lab, Department of Electrical and Computer Engineering

- Developed a **cross-layer co-design methodology** for photonic integrated circuit (PIC) based AI accelerators spanning device physics, circuit design, system architecture, and ML algorithm evaluation
- Designed and validated multiple **microring-resonator (MRR) based photonic accelerator architectures** for CNN inference, addressing fundamental challenges in scalability, energy efficiency, and hardware utilization
- Developed a transaction-level, event-driven simulator to evaluate the performance of photonic AI accelerators
- Published **20+ peer-reviewed papers** in premier venues (IEEE TCAD, ACM TODAES, IPDPS, NOCS)

## Software Engineer

Jul 2016 – Jul 2019

EPAM Systems Pvt. Ltd., Telangana, India

- Architected Java-based backend services with PostgreSQL integration for enterprise document management platforms
- Developed intelligent document processing pipelines leveraging WorkFusion NLP automation, reducing manual processing time by 60%
- Implemented CI/CD pipelines, comprehensive unit testing, and Agile development practices using JIRA and Git

## Research Funding

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- **Co-PI**, “Simulation-Driven Co-Design of Multidimensional Photonic Computing Systems (COSMOS),” BNL LDRD Type B, Oct 2026 – Sep 2028 (PI: P. Baity). One of 12 proposals awarded laboratory-wide and the sole selection from the department

## Key Technical Contributions

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### Ph.D. Research Highlights

- **Reconfigurable Photonic CNN Accelerators:** Designed architectures supporting mixed-size tensors and diverse layer configurations, reducing hardware underutilization by 40% and improving throughput  $3.2\times$  over fixed designs
- **Hybrid Time-Amplitude Encoded Optical Accelerator (HEANA):** Developed flexible dataflow architectures for energy-efficient GEMM and CNN inference, achieving  $5\times$  energy reduction versus digital baselines

- **Stochastic Computing Photonic Accelerator (SCONNA):** Enabled ultra-fast inference of integer-quantized CNNs with improved scalability-precision trade-offs, demonstrating 10× speedup with 12% accuracy loss
- **Electro-Photonic Logic Primitives:** Created unary MAC and XNOR-bitcount accelerators for binary neural networks, achieving 100× energy efficiency improvements
- **Photonic Network-on-Chip Framework (PROTEUS):** Developed a loss-aware self-adaptation framework co-managing laser power and communication performance in multi-core photonic systems
- **Hardware Security in Photonic NoCs:** Exploited process variations for protection against snooping and hardware attacks, demonstrating novel security-by-physics approaches

## Selected Publications

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### Journal Articles (Recent)

1. [Authors incl. **S. S. Vatsavai**], “A System-level Exploration of Superconducting Supercomputers for Emerging Workloads,” *IEEE Transactions on Computers*, 2026 (accepted).
2. [Authors incl. **S. S. Vatsavai**], “Predicting Job Turnaround Time in Large-Scale Distributed Computing Environments with Graph Neural Networks,” *EPJ Research Infrastructures*, vol. 10, art. 14, 2026.
3. [Authors incl. **S. S. Vatsavai**], “MCP Agents in PanDA,” *Journal of Instrumentation*, vol. 21, no. 07, C07030, 2026.
4. T. Chowdhury, T. Maeno, F. F. Akman, **S. S. Vatsavai**, et al., “Machine Learning-Driven Predictive Resource Management in Complex Science Workflows,” *International Journal of Modern Physics A*, vol. 41, no. 03, 2650025, 2026.

### Conference and Workshop Papers (Recent)

1. [Authors incl. **S. S. Vatsavai**], “CGSim: A Simulation Framework for Large Scale Distributed Computing Infrastructures with AI-Driven Applications,” *22nd IEEE International Conference on eScience*, Naples, Italy, 2026.
2. [Authors incl. **S. S. Vatsavai**], “LightshowAI-Agent: An MCP-Based Agentic Framework for Real-Time and Reproducible X-Ray Absorption Spectral Analysis at Scientific User Facilities,” *AI4S Workshop, SC’26*, 2026.
3. [Authors incl. **S. S. Vatsavai**], “HOLOS: A Gradient-Descent-Based Design-Space Exploration Framework for Hierarchical AI Accelerator Systems,” *Proc. SC’26 Workshops*, 2026.
4. [Authors incl. **S. S. Vatsavai**], “DMSim: Configurable Data Management Simulation Framework for Scientific Computing Grid,” *Proc. SC’26 Workshops*, 2026.
5. [Authors incl. **S. S. Vatsavai**], “HorizonGuard: Measuring Hindsight Contamination in Generative AI Cyber Threat Intelligence,” *IEEE Generative AI for Secure Systems (GAISS)*, 2026.
6. [Authors incl. **S. S. Vatsavai**], “ChronoEntity: Auditable Agentic AI for Point-in-Time Entity Integrity in Financial Data Pipelines,” *IEEE Generative AI for Secure Systems (GAISS)*, 2026.
7. **S. S. Vatsavai**, R. K. Ahmed, K. C. Hsu, et al., “CGSim: A Simulation Framework for Large-Scale Distributed Computing Environment,” *Proc. SC’25 Workshops (PMBS)*, 2025. **Best Short Paper Award**
8. S. Dutta, O. O. Kilic, **S. S. Vatsavai**, et al., “Error Analysis of Globally Distributed Workflow Management System,” *Proc. SC’25 Workshops*, 2025.
9. F. F. Akman, S. Yoo, A. Hoisie, **S. S. Vatsavai**, et al., “Simulating the ATLAS Distributed Computing Infrastructure to Develop and Optimize Workload Allocation Strategies,” *ATL-COM-SOFT-2025-045*, 2025.

## Under Review and Preprints

1. **S. S. Vatsavai**, et al., “TRASSE: Learned Simulator Surrogates for Scheduling-Policy Search at Distributed Scientific Computing Facilities,” *under review*.
2. O. O. Kilic, D. K. Park, Y. Ren, **S. S. Vatsavai**, et al., “Towards an Introspective Dynamic Model of Globally Distributed Computing Infrastructures,” *arXiv:2506.19578*, 2025.
3. S. Feng, J. Kim, Y. Yang, **S. S. Vatsavai**, et al., “Alternative Mixed Integer Linear Programming Optimization for Joint Job Scheduling and Data Allocation in Grid Computing,” *arXiv:2502.00261*, 2025.

## Photonic & Emerging AI Accelerators

1. **S. S. Vatsavai**, V. S. P. Karempudi, I. Thakkar, “HEANA: A Hybrid Time-Amplitude Analog Optical Accelerator with Flexible Dataflows for Energy-Efficient CNN Inference,” *ACM Trans. Design Automation of Electronic Systems (TODAES)*, 2025.
2. **S. S. Vatsavai**, I. G. Thakkar, “Photonic Reconfigurable Accelerators for Efficient Inference of CNNs with Mixed-Sized Tensors,” *IEEE Trans. Computer-Aided Design (TCAD)*, vol. 41, no. 11, pp. 4337–4348, 2022.
3. **S. S. Vatsavai**, V. S. P. Karempudi, I. Thakkar, T. Hastings, A. Salehi, “SCONNA: A Stochastic Computing Based Optical Accelerator for Ultra-Fast, Energy-Efficient Inference of Integer-Quantized CNNs,” *IEEE IPDPS*, 2023.
4. **S. S. Vatsavai**, V. S. P. Karempudi, I. Thakkar, “PROTEUS: Rule-Based Self-Adaptation in Photonic NoCs for Loss-Aware Co-Management of Laser Power and Performance,” *IEEE/ACM NOCS*, 2020.
5. O. A. Alo, **S. S. Vatsavai**, I. Thakkar, “Scaling Analog Photonic Accelerators for Byte-Size, Integer General Matrix Multiply (GEMM) Kernels,” *IEEE ISVLSI*, 2024.
6. C. Ti, U. Hassan, **S. S. Vatsavai**, et al., “Mixed Delay/Nondelay Embeddings Based Neuromorphic Computing with Patterned Nanomagnet Arrays,” *IEEE ICRC*, 2024.

Full publication list available on [Google Scholar](#)

## Invited Talks and Presentations

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- **Lightning Talk and Poster**, “AI-Enabled ModSim for Distributed Computing: A Simulator-to-Surrogate Methodology for Scalable Performance Modeling,” ModSim 2026, Seattle, WA, Aug 2026
- **Plenary Talk**, “Resilient Federated Workflows in a Heterogeneous Computing Environment: The REDWOOD Project,” ATLAS Software and Computing Week, CERN, Geneva, Switzerland, Jul 2026
- **Talk and Poster**, REDWOOD project progress and future directions (on behalf of PI A. Klimentov), DOE ASCR Computer Science PI Meeting, Jul 2026
- **Mini-Symposium Talk and Panelist**, “AI-Enabled Modeling, Simulation, and Optimization of Distributed Computing Systems,” PASC 2026, University of Bern, Switzerland, Jun 2026
- **Invited Talk**, “Cross-Layer Design of Highly Scalable and Energy-Efficient AI Accelerator Systems Using Photonic Integrated Circuits,” 4th Innovations in Optics, Photonics, and Laser Engineering Conference (Optics-Laser 2026), Boston, MA, May 2026
- **Live Demonstration**, real-time X-ray absorption spectroscopy analysis with AmSC-enabled LightshowAI, Confab26, Rockville, MD, May 2026 (featured on the AmSC website)

## Honors and Awards

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- Best Short Paper Award, PMBS Workshop @ SC’25 (2025)

- Shining Star Award, University of Kentucky Center for Graduate and Professional Diversity Initiatives (May 2024)
- Best Doctoral Student Award, ECE Research Symposium, University of Kentucky (May 2023)

## Technical Skills

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**Programming & Development:** Python, C++, Java, JavaScript, Bash, CMake, Git/GitLab CI/CD, Docker, Kaniko

**AI/ML Frameworks:** PyTorch, TensorFlow, Keras, scikit-learn, PyG, MLflow, model quantization, surrogate modeling, transformer architectures, graph neural networks

**Agentic AI:** Model Context Protocol (MCP) servers and tools, Chainlit, LLM agent integration

**HPC & Simulation:** SimGrid, GEM5, Calculon, Timeloop, STONNE, SPEC2017, hierarchical performance/energy modeling, analytical modeling, distributed systems simulation

**Web & Data Infrastructure:** Dash, Plotly, Flask, RESTful APIs, OIDC/SSO (Authlib), Apache reverse proxy, Tiled, Materials Project API

**Photonics Tools:** Ansys Lumerical (MODE, CHARGE, INTERCONNECT), photonic integrated circuit design and simulation

**Data Analysis:** NumPy, Pandas, Matplotlib, data preprocessing pipelines, large-scale dataset analysis

## Professional Service & Activities

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### Peer Review Activities (40+ papers reviewed)

- **Journals:** IEEE Transactions on Computers, IEEE Transactions on Computer-Aided Design (TCAD), IEEE Transactions on Sustainable Computing, IEEE Access, Optical and Quantum Electronics, Memories (Elsevier)
- **Conferences:** SC (2025, 2026), VLSID (2023–2026), GLSVLSI (2024–2026), ISVLSI 2024, ISPED 2023, CODES+ISSS 2023, ICCD

### Volunteer Activities

- Open House Volunteer, Brookhaven National Laboratory
- Science Bowl Volunteer, Brookhaven National Laboratory